

NASA WALLOPS, VA, USA

WMO: 724020

Lat: 37.937N

Lon: 75.466W

Elev: 14

StdP: 101.16

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 93739

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.2	(c) -5.9	(d) -17.4	(e) 0.8	(f) -6.6	(g) -14.7	(h) 1.0	(i) -4.2	(j) 12.3	(k) 4.0	(l) 11.1	(m) 3.8	(n) 4.0	(o) 320	(p) 0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	32.9	24.9	31.5	24.4	30.2	23.9	26.4	30.4	25.7	29.3	25.2	28.5	4.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.3	20.6	28.4	24.7	19.7	27.8	24.1	19.0	27.2	82.1	30.3	79.1	29.3	76.6	28.6	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	9.3	8.4	-11.8	35.2	3.0	1.5	-14.0	36.2	-15.7	37.1	-17.4	38.0	-19.5	39.0
			-12.7	27.6	2.7	0.8	-14.7	28.2	-16.3	28.7	-17.8	29.1	-19.8	29.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	3.1	4.3	7.7	12.9	18.0	23.0	25.5	24.9	21.7	15.9	9.8	5.8
	DBStd	8.93	5.57	4.82	4.74	4.26	4.02	3.42	2.66	2.49	3.23	4.35	4.49	4.88
	HDD10.0	722	221	168	104	17	0	0	0	0	0	4	58	150
	HDD18.3	2186	471	392	331	170	58	5	0	0	9	102	257	391
	CDD10.0	2341	9	9	32	105	247	392	482	461	351	186	52	18
	CDD18.3	763	0	0	1	8	46	147	223	202	109	26	1	1
	CDH23.3	5838	0	1	5	53	270	1085	2027	1662	630	103	1	1
	CDH26.7	1714	0	0	0	9	61	332	699	476	118	18	0	0
Wind	WSAvg	4.1	4.2	4.3	4.7	4.8	4.3	4.0	3.7	3.5	4.0	3.9	3.8	3.8
Precipitation	PrecAvg	1011	83	76	103	74	84	76	98	101	83	72	66	86
	PrecMax	1490	186	154	174	128	205	193	307	332	248	174	188	222
	PrecMin	741	22	19	8	10	11	24	7	14	17	0	1	13
	PrecStd	193	40	33	40	29	50	41	59	71	55	42	43	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.7	20.4	23.6	28.0	30.7	34.0	35.5	33.9	31.6	28.9	22.4	21.2
		MCWB	15.3	15.5	16.4	18.7	22.2	24.4	25.3	25.7	24.8	22.3	18.0	16.9
	2%	DB	16.0	16.6	19.8	24.6	28.4	32.1	33.3	32.0	29.5	25.9	20.1	17.5
		MCWB	12.8	12.4	14.0	17.9	21.6	24.0	25.1	24.9	23.6	21.0	16.7	14.9
	5%	DB	13.7	13.7	17.1	21.9	26.3	30.2	31.7	30.6	27.9	24.0	18.4	15.3
		MCWB	11.3	10.6	12.7	16.4	20.5	23.3	24.7	24.3	23.0	19.9	15.5	13.4
	10%	DB	11.5	11.5	14.7	19.5	24.4	28.4	30.2	29.3	26.6	22.4	16.8	13.4
		MCWB	9.5	8.9	11.3	14.9	19.5	22.6	24.2	23.9	22.3	19.0	14.4	11.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.7	16.5	17.7	21.1	23.9	26.1	27.5	27.1	26.3	23.8	19.8	18.1
		MCDB	18.5	19.6	22.0	25.5	28.4	30.4	32.6	31.1	29.5	26.5	21.2	20.4
	2%	WB	13.7	13.4	15.3	18.9	22.6	25.2	26.6	26.3	25.2	22.3	17.8	15.7
		MCDB	15.4	15.2	18.7	23.0	26.7	29.7	30.9	30.1	27.6	24.9	19.2	17.1
	5%	WB	11.9	11.3	13.8	17.3	21.5	24.5	25.8	25.6	24.4	21.1	16.4	13.8
		MCDB	13.1	13.0	16.1	20.8	25.1	28.5	29.7	28.9	26.7	23.3	17.9	14.9
	10%	WB	9.8	9.3	11.9	15.9	20.4	23.8	25.2	25.0	23.6	19.9	14.9	11.8
		MCDB	11.2	11.2	14.1	18.8	23.7	27.2	28.8	28.0	25.8	21.7	16.4	13.2
Mean Daily Temperature Range	5% DB	MDBR	8.5	8.8	8.9	9.2	8.7	8.1	7.9	7.8	8.0	8.8	9.0	8.8
		MCDBR	10.7	11.7	12.0	12.3	10.9	9.8	9.3	8.6	8.4	9.2	9.7	10.4
		MCWBR	8.8	9.0	8.1	6.5	5.5	4.3	3.9	3.8	4.4	5.4	7.4	8.7
	5% WB	MCDBR	10.2	10.8	10.8	10.7	9.8	8.5	8.0	7.5	7.0	7.6	8.5	9.7
		MCWBR	9.2	9.5	8.4	6.7	5.7	4.2	3.9	3.6	4.0	5.1	7.5	8.9
Clear-Sky Solar Irradiance	taub		0.317	0.323	0.352	0.393	0.439	0.481	0.508	0.502	0.409	0.361	0.329	0.318
	taud		2.510	2.488	2.441	2.340	2.245	2.177	2.117	2.120	2.367	2.497	2.534	2.540
	Ebn at Noon		868	909	910	886	847	809	783	778	839	854	848	842
	Edh at Noon		79	91	105	123	138	148	156	152	112	90	77	72
All-Sky Solar Radiation	RadAvg		2.21	3.02	4.10	5.23	5.90	6.43	6.21	5.59	4.56	3.48	2.50	1.95
	RadStd		0.19	0.31	0.33	0.36	0.59	0.45	0.29	0.34	0.41	0.34	0.21	0.17

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.64	N/A	N/A	N/A	N/A	N/A	N/A	-121	+239	+133
(47) Regional (68 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+152	+96

Nomenclature: See separate page